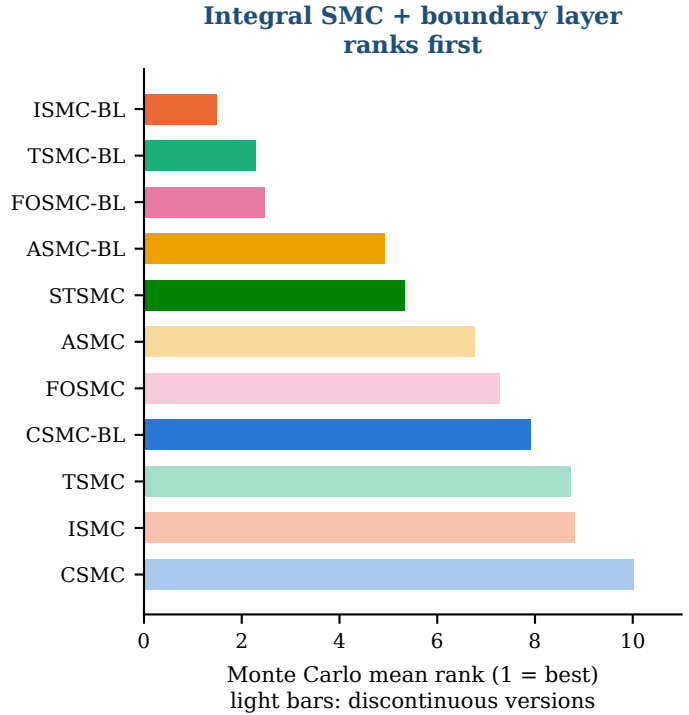
Plant & drive

Siemens 1FK7063 SPMSM (data sheet), PI current loops, $V_{dc}/\sqrt{3}$ and $I_{\max}$ limits, delay, 20-bit encoder, T_s=100 µs

Six SMC laws

CSMC · ISMC · TSMC
ASMC · FOSMC · STSMC
sgn and boundary layer
no load in control law
Lyapunov gain bounds

Fair evaluation

identical DE tuning
(1920 evals, 3 seeds)
7 unseen scenarios
200-run Monte Carlo
Friedman test